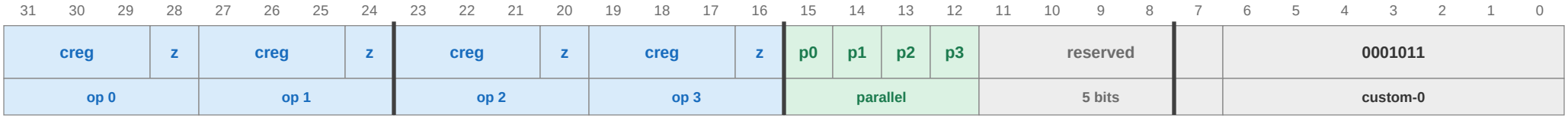
Starbug 2 — Execute-Packet Header

custom-0 major opcode 0001011 (0x0B) | 25 usable bits, 31:7



Fields

creg / z, bits 31:16 — Four nibbles, one per operation, in the C6x positions (31:28, 27:24, 23:20, 19:16). creg = 0 with z = 0 means unconditional. creg selects one of x9–x15, giving 7 predicate registers; z = 0 tests nonzero, z = 1 tests zero.

p bits, 15:12 — pi = 1 means operation i+1 issues in parallel with operation i. Only three boundaries exist among four operations, so p3 is spare.

reserved, 11:7 — Freed when autoincrement moved into the custom-1 load/store format. Reserve for a wider slot count: 8 slots would need 32 bits of creg/z alone and would force a second header word, as the C6x needed two extension slots.

opcode, 6:0 — custom-0. inst[1:0] = 11 makes it a 32-bit instruction; an implementation without the extension raises an illegal-instruction exception rather than ignoring it, which is why this replaced the 16-bit HINT header.

Packet semantics

- The header's presence makes the following operations an execute packet, so there is no mode: entry is a branch to a header, and exit is simply the absence of one.
- Operation lengths are not encoded here — each operation declares its own width in its inst[1:0], so the decoder walks the packet.
- All operations in a packet read their sources at the start of the cycle and commit their writes at the end.
- At most one memory operation per packet (single load/store data path).
- Two predicated operations may write the same destination when their predicates are complementary (dynamic rule).